

Adaptive Advance-Time Control for nFAPI Split under Unpredictable Latency

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Abstract—

Index Terms—nFAPI, O-RAN, Functional Split, Adaptive Control.

I. Introduction

A. Introduction

In recent years, the Open Radio Access Network (O-RAN) architecture has become popular. O-RAN allows for more flexible network deployments. The Network Functional Application Platform Interface (nFAPI) is an important part of this architecture. nFAPI helps split the O-RAN Distributed Unit (O-DU) into O-DU High and O-DU Low parts. This functional split separates the Media Access Control (MAC) layer and the Physical (PHY) layer. Figure 1 shows the O-RAN nFAPI split architecture.

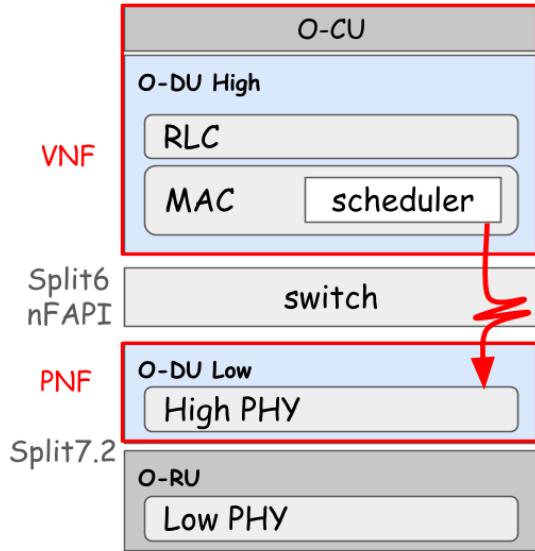


Fig. 1. O-RAN nFAPI functional split architecture.

According to the nFAPI specification, the Virtual Network Function (VNF) includes Layer 2 and Layer 3 operations. The Physical Network Function (PNF) handles the Layer 1 Physical layer processing. Deploying the VNF and PNF on different machines causes several challenges. Strict slot synchronization is required between these distributed components. Different traffic profiles affect the latency between the VNF

and PNF. Figure 2 shows the relationship between different traffic types and the VNF-PNF latency.

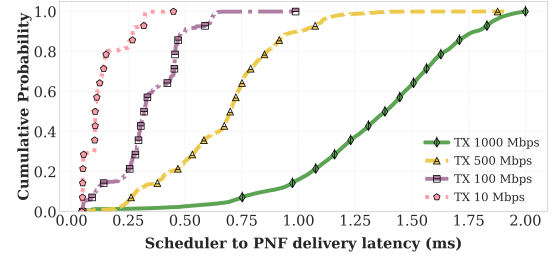


Fig. 2. Observation of latency between VNF scheduler and PNF under different traffic loads.

To lower deployment costs, operators often want to deploy Split-6 nFAPI gNBs over non-ideal networks. However, unpredictable network latency in these environments introduces significant challenges for MAC scheduler timing. Figure 3 shows the latency jitter in a non-ideal deployment. Existing open-source solutions, such as OpenAirInterface (OAI), provide basic nFAPI support. However, they do not fully comply with the nFAPI specification for delay management. Therefore, they cannot handle the unpredictable latency challenges. To address these issues, this paper proposes a dynamic trigger scheduler adjustment algorithm. Our proposed algorithm fully complies with the specification. We verify the effectiveness and benefits of this dynamic adjustment through systematic experiments.

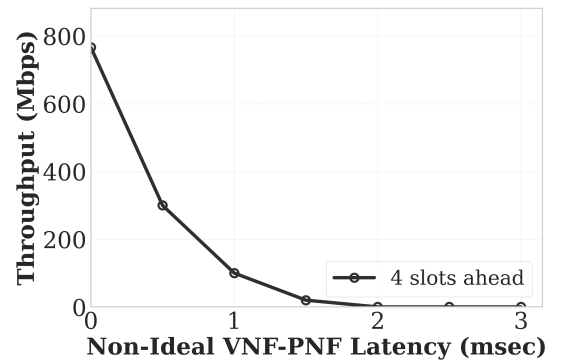


Fig. 3. Impact of unpredictable network latency in a non-ideal environment.

B. Related Works

C. Contributions

The main contributions of this paper are summarized as follows.

- Contribution 1.
- Contribution 2.
- Contribution 3.

The rest of the paper is organized as follows. Section II defines the system model/architecture considered in this paper. The proposed analytical model/method is given in Section III. Section IV shows the numerical results. Concluding remarks are given in Section VI.

II. System Model/Architecture

TABLE I
nFAP Delay Management Configuration Parameters

Parameter	Type	Description
timingWindow	uint16_t	Timing window for delay management. See section 3.2.9 of [10]
timingMode	uint8_t	Timing mode for Timing Info. See section 3.2.9 of [10]
timingPeriod	uint8_t	Timing period for Timing Info. See section 3.2.9 of [10]

The following assumptions were made in this paper.

- Assumption 1.
- Assumption 2.
- Assumption 3.

III. Proposed Method

IV. Numerical/Simulation/Experimental Results

Computer simulations/Experiments were conducted to verify xxx (the effectiveness of the proposed analytical model). In the following figures, each point represented the average value of 10^5 samples. In each sample, we collect the statistics of xxx. Symbols and lines were used to show the simulation and analytical results, respectively.

XXX scenarios were investigated. Scenario I was designed to demonstrate/verify the accuracy/effectiveness of Contribution 1.

A. Contribution 1

Fig. 1.1: Purpose X axis: Y axis:

B. Contribution 2

Fig. 2.1: X axis: Y axis:

C. Contribution 3

Fig. 3.1: X axis: Y axis:

References